

HARIS WAHEED BHATTI

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PROFESSIONAL SUMMARY

BIM and digital construction specialist completing a European Master's in Building Information Modelling at the University of Ljubljana, building on a Bachelor's in Civil Engineering from NUST, Pakistan. Work spans IFC-to-Neo4j knowledge graph pipelines with LLM-based retrieval (built with Claude), and blockchain-based construction document management. Based in Ljubljana, Slovenia with full-time work rights under a Slovenian residence permit; open to BIM Software Developer, BIM Manager, Information Manager, and Digital Construction Specialist roles.

TECHNICAL SKILLS

BIM & AEC Tools: IFC, Revit, Dynamo, Blender, Autodesk Construction Cloud, Three.js, WebIFC

Programming & Data: TypeScript, Python, Solidity, Neo4j, Knowledge Graphs

AI & LLM: Claude, Retrieval-Augmented Generation (RAG), Artificial Intelligence

Languages: Urdu (Native), English (Fluent)

EXPERIENCE

Contract Management Intern 07/08/2023 – 01/09/2023
Engineering and Management Services Pakistan

- Reviewed construction contract reports and arbitration case documentation under **FIDIC** conditions, and presented findings on FIDIC and arbitration procedures to the internal legal team

Research Intern (Part-time, Remote – Concurrent with Above) 17/07/2023 – 24/08/2023
Southern Illinois University Edwardsville USA

- Extracted and synthesized findings from academic articles and abstracts to support a faculty-mentored research proposal, applying feedback to refine presentation and analysis technique | Supervisor: linkedin.com/in/saadullah1

Civil Engineering Intern 11/07/2023 – 04/08/2023
Fatima Fertilizer Company Pakistan

- Modeled a 42m x 162m (6,804 m²) rock phosphate storage shed independently in 3D from 2D drawings, coordinating with stakeholders to refine the design

PROJECTS

IFC Knowledge Graph & RAG Pipeline | Neo4j, LLMs, IFC, Claude | Master's Thesis 2026

- Modeling IFC building data as a property graph in **Neo4j**, preserving hierarchical and relational structures of BIM entities
- Implementing a **RAG architecture** combining LLMs and graph retrieval to enable semantic querying and knowledge discovery within BIM datasets, built with **Claude** for AI-assisted development
- Building an automated pipeline for IFC ingestion and graph serialization, supporting bidirectional data flow between BIM files and graph environments

Pakistani-Themed Expo BIM Management | IFC, Autodesk Construction Cloud 2024

- Spearheaded a BIM management project for a Pakistani-themed expo in Guimarães, Portugal, leveraging **IFC** and **Autodesk Construction Cloud** to develop a comprehensive technical design
- Facilitated collaboration between Architecture and MEP disciplines, creating a **federated model** that integrated diverse design inputs; conducted **clash detection** and compliance checks against all information requirements

Blockchain-Based BIM Document Management | Solidity | Bachelor's Thesis 2024

- Developed a **Solidity** smart contract to digitally sign and manage construction documents, with off-chain storage linked to the blockchain for immutability and transparency, funded by a **PKR 100,000 grant** from the Pakistan Engineering Council | [Contract](#)

EDUCATION

University of Ljubljana Slovenia
European Master in Building Information Modelling 15/11/2025 – 30/09/2026

- Field of study: Information Management in Construction. Thesis: IFC-to-Neo4j knowledge graph with LLM-based RAG pipeline (see Projects)

National University of Science and Technology Pakistan
BE Civil Engineering 09/2020 – 06/2024

- Final grade: 3.25/4.00. Thesis: Blockchain-based BIM document management (see Projects)

ACHIEVEMENTS & AWARDS

Pakistan Engineering Council Grant (PKR 100,000)	– Awarded for final-year blockchain-based BIM thesis project	2024
Director, School Orientation Committee	– Coordinated logistics and onboarding programming for incoming student cohorts, NUST	2020–2024
President, NUST 3v3 Madness Basketball Tournament	– Organized team registration and event logistics for a university-wide competition	2020–2024
Coursera: Supervised Machine Learning – Regression and Classification	– DeepLearning.AI	2023
Coursera: Management of Urban Infrastructures – Part 1	– EPFL	2024
Coursera: Innovative Governance of Large Urban Systems		2024